

Homework (Jalapeno)

- Q1.** The ancient Egyptians built the Great Pyramid using $2x \cdot 3y$ stones. If they wanted to factor out the greatest common factor, what would it be?
- (A) 2
(B) 3
(C) 6
(D) x
(E) y
- Q2.** The ancient Greek city of Babylon had $12x + 16$ olive trees. What is the greatest common factor of this expression?
- (A) 2
(B) 4
(C) 8
(D) 12
(E) 16
- Q3.** The Mayan civilization used $9x^2 + 27x$ to calculate the area of their temples. What is the factored form of this expression?
- (A) $3x(3x + 9)$
(B) $9x(x + 3)$
(C) $3(3x^2 + 9x)$
(D) $9(x^2 + 3x)$
(E) $3x(3x - 9)$
- Q4.** The ancient Chinese used $24x - 30$ to measure the length of the Great Wall. What is the greatest common factor of this expression?
- (A) 2
(B) 6
(C) 12
(D) 24
(E) 30
- Q5.** The Inca Empire used $15x + 25$ to calculate the amount of gold they had. What is the greatest common factor of this expression?
- (A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q6.** The ancient Indians used $8x^2 - 24x$ to calculate the area of their crops. What is the factored form of this expression?
- (A) $8x(x - 3)$
(B) $8(x^2 - 3x)$
(C) $8x(x + 3)$
(D) $8(x^2 + 3x)$
(E) $8x(3x - 1)$
- Q7.** The ancient Egyptians used $6x + 18$ to measure the length of their pyramids. What is the greatest common factor of this expression?
- (A) 2
(B) 3
(C) 6
(D) 12
(E) 18
- Q8.** The ancient Greeks used $20x - 12$ to calculate the amount of olive oil they produced. What is the greatest common factor of this expression?
- (A) 2
(B) 4
(C) 8
(D) 12
(E) 20

Open Ended Questions (FRQ)

- Q9.** The ancient Mesopotamians used $12x^2 + 36x$ to calculate the area of their cities. Factor out the greatest common factor and simplify the expression. Show your work.
- Q10.** The ancient Aztecs used $15x + 30$ to calculate the amount of food they had. Factor out the greatest common factor and simplify the expression. Explain your reasoning.

Chef's Tips

- Tip 1: Make sure students show their work for the free response questions.
- Tip 2: Encourage students to use different methods to factor out the greatest common factor.
- Tip 3: Remind students to check their answers for accuracy.